Run report: `research/la_engines_20261014`, 2026-10-14T17:28Z – 2026-10-15T01:14Z

Descriptive record of every run; interpretation is in `../../REPORT.pdf`.

Commands. `scripts/grid.sh` (frozen protocol, `PROTOCOL.md`, commit `404251aee` frozen, instrument fixes `8a82cf286` before data), `scripts/grid_amend1.sh` (`abdf48826`), `scripts/grid_amend2.sh` (`5731279cf`), `scripts/rerun_contended.sh` (amendment 3), `scripts/grid_amend4.sh` (`dca78f544`), `scripts/grid_amend5.sh` (`76f86d667`), `scripts/e2e.sh` (`ic logs`, configurations `n23-legacy:23:2:0:legacy`, `n31-search:31:3:0:search`, `n37-search:37:3:0:search`). CADO-NFS `692ecb7e62f0f3bdab88ee44cc60b8ded0ea1a1b`, Release, GF(p) layers `p1;p3`.

Host. 4 vCPU Intel Xeon @ 2.80 GHz (AVX-512), 15 GiB, Linux 6.18.44, rustc 1.97.0; shared cloud VM, no core reservation: isolation L0. Rayon pool 4 threads (1 in the one-thread pass); CADO `thr=2x2` (`1x1`).

Workload. Synthetic relation systems, m = 3 (m = 2, 4 at D = 16,384), D = 2^{10}…2^{18}, seeds 1–3, modulus $2^{62} - 57$; coefficients ±1, or +1 for amendments 1, 2 and the `-pos` arms. Real: `ic logs` on Koblitz curves, seeds 1–3, three linear-algebra modes.

records	runs	failed	verified	timeouts	file
frozen grid	210	4	192	3	`raw/grid.jsonl`
amendment 1	72	1	70	0	`raw/grid_amend1.jsonl`
amendment 2	57	0	57	0	`raw/grid_amend2.jsonl`
amendment 3 re-runs	38	1	35	0	`raw/rerun_contended.jsonl`
amendment 4	243	0	242	0	`raw/grid_amend4.jsonl`
amendment 5	135	0	134	0	`raw/grid_amend5.jsonl`
`ic logs` end to end	27	0	27	0	`raw/e2e.jsonl`

Failures and unverified runs, with their recorded reasons. Timeouts (1,800 s): `core-bw`, `core-bw` A/A and `cado-bwc` at D = 262,144 seed 1 (the stop rule then skipped seeds 2–3). `cado-bwc` m = 2 seed 1: exit 1 on a 25-column core. Amendment 1 `cado-full` D = 32,768 seed 3 and its re-run: `bwc` balancing abort ("expects a column permutation replicated on rows"); the first amendment-1 start used a stale binary (`raw/grid_amend1.falsestart.*`). Unverified: frozen-grid `core-bw`/`cado-bwc` on three singular unmerged cores (D = 8,192 s3, 16,384 s3, 32,768 s1; CADO `rows_ok` true); CADO-NFS pipeline D = 1,024 seed 2 in every session.

Measured values. Medians and min–max per (arm, D) for every pass are in `../../summary.json` (computed by `scripts/plots.py` from these records, no other arithmetic); per-run values, CADO per-step wall times (`steps_wct`) and filter reports are in the records.

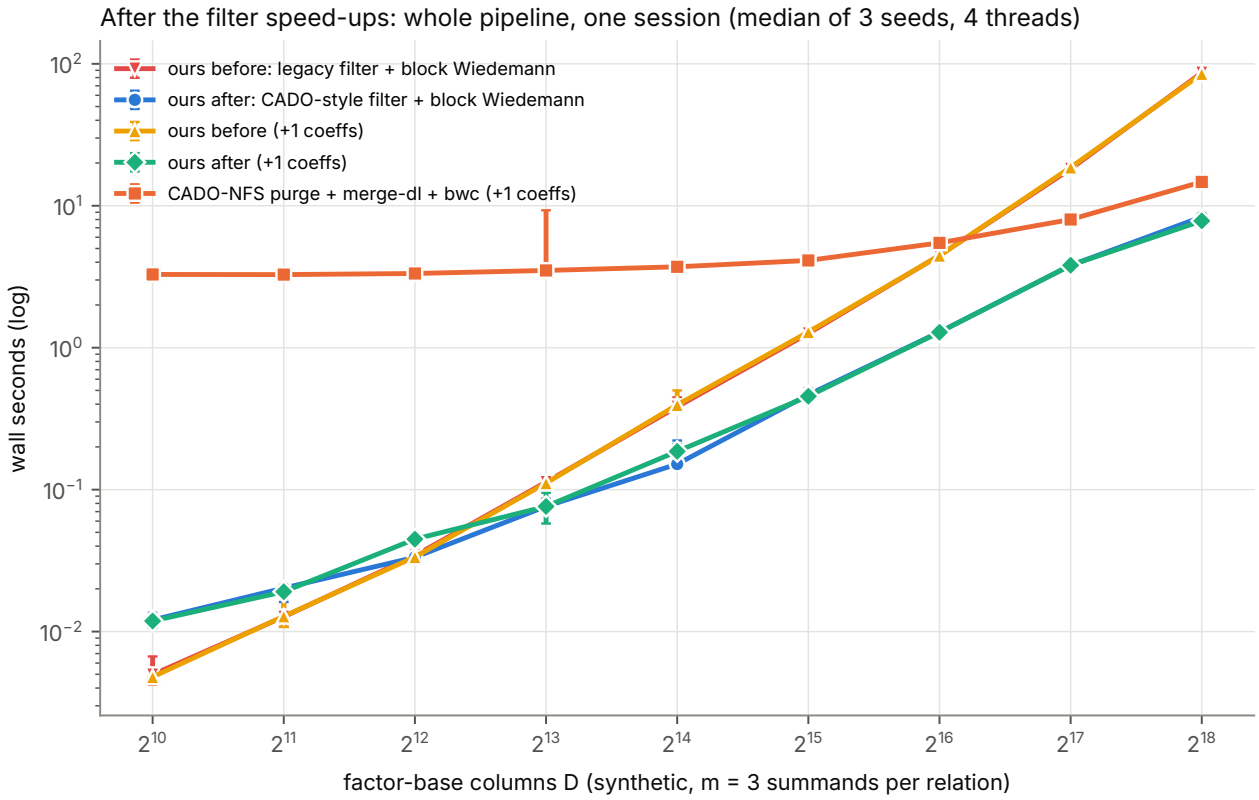


Figure 1: Amendment-5 session (records `raw/grid_amend5.jsonl`), median of 3 seeds, bars min–max, `solve_s`.

Controls. Every arm is checked against the planted logarithm vector. A/A: `core-bw` repeated on seed 1 at every D (frozen grid). Contention: 38 runs overlapping builds or diagnostics re-run on the idle host (amendment 3).